**CPPC 18/L**

COMPUTER NETWORK & SECURITY

**ACTIVITY No.5**

**DHCP IPv6**

**Submitted by:**

**LOGATOC, APRIL FOILA JOY M.**

BS CpE 3B

**Submitted to:**

**ENGR. JAN ERROL MAMPUSTI**

Subject Professor

## I. Objectives

- To enable the DHCPv6 service on a server.
- To allow computers to automatically receive IPv6 addresses.
- To understand how DHCPv6 helps in managing IPv6 address assignment.
- To configure a simple network using a server, switch, and PCs in Cisco Packet Tracer.

## II. Materials

- Cisco Packet Tracer
- 1 Server
- 1 Switch
- 2–3 PCs
- Laptop
- Copper Straight-Through cables

## III. Procedures

1. Create the Network Topology
   - Open Cisco Packet Tracer.
   - Add the needed devices in the workspace:
     - 1 Server
     - 1 Switch
     - 2–3 PCs
   - Connect all devices using Copper Straight-Through cables.
   - Connect the server to the switch.
   - Connect each PC to the same switch.
   - Make sure that all connections are active and properly linked.

2. Configure the Server IPv6 Address
   - Click the Server.
   - Go to Desktop.
   - Open IP Configuration.
   - Under the IPv6 Configuration section, set the following:
     - IPv6 Address: 2001:db8:1::10
     - Prefix Length: 64
   - This IPv6 address will serve as the server's address within the network.

3. Enable the DHCPv6 Service
   - Click the Server again.
   - Go to the Services tab.
   - Select DHCPv6.
   - Turn the DHCPv6 service ON.
   - This enables the server to automatically provide IPv6 addresses to the connected PCs.

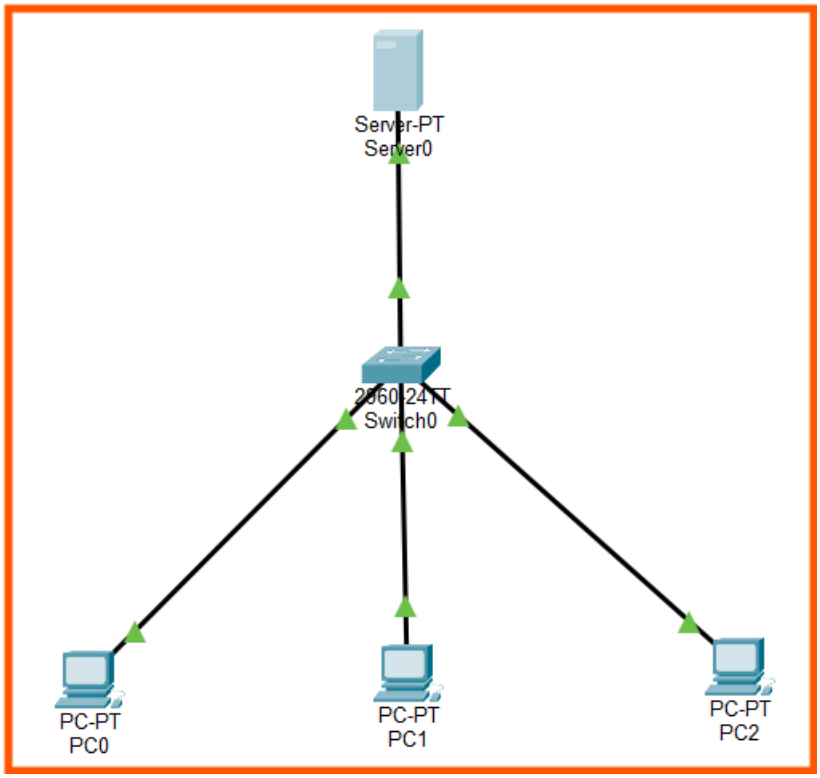
4. Configure the DHCPv6 Pool
   - In the DHCPv6 settings, create a new DHCPv6 pool.
   - Enter the following details:
     - Pool Name: IPv6Pool
     - Prefix: 2001:db8:1::
     - Prefix Length: 64
     - DNS Server: ::
   - Click Add to save the DHCPv6 pool.
   - This pool will be used by the server to assign IPv6 addresses to the client computers.

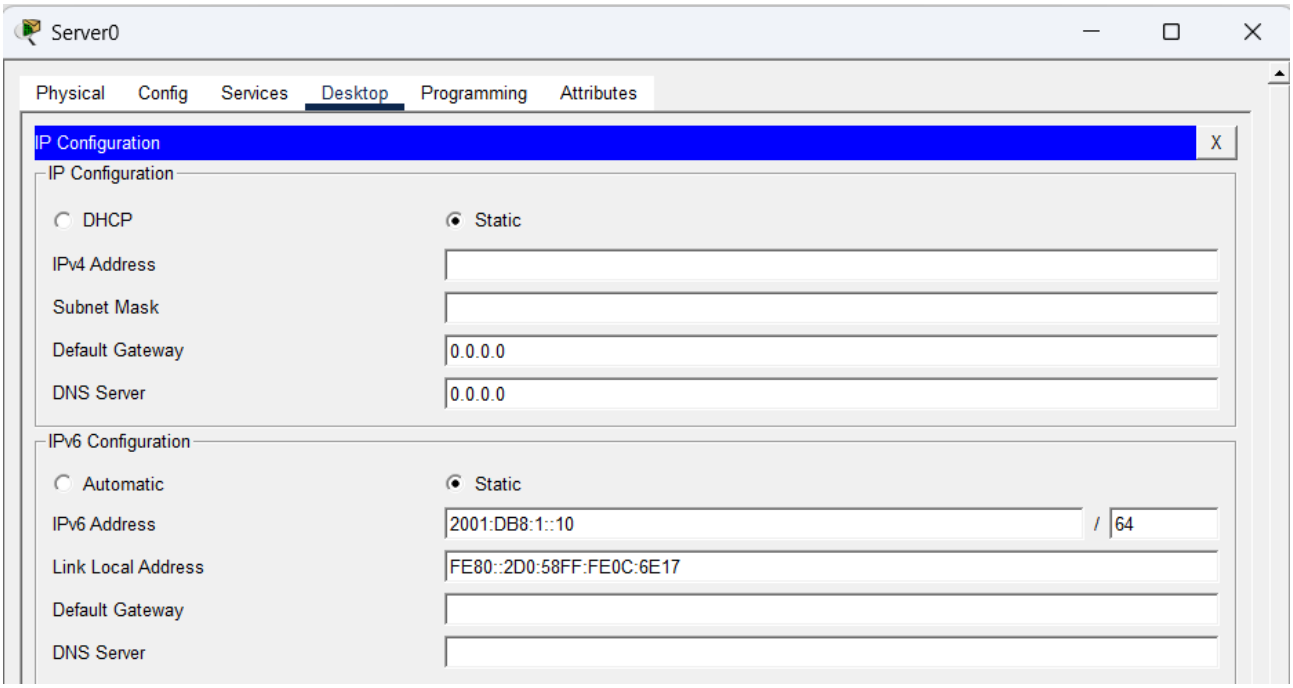
5. Set the PCs to Automatic IPv6
   - Click each PC one by one.
   - Go to Desktop.
   - Open IP Configuration.
   - Under the IPv6 Configuration section, select Automatic.
   - Wait until the IPv6 address is assigned successfully.
   - Repeat the same process for the other PCs.
   - Each PC should receive an IPv6 address from the DHCPv6 server.

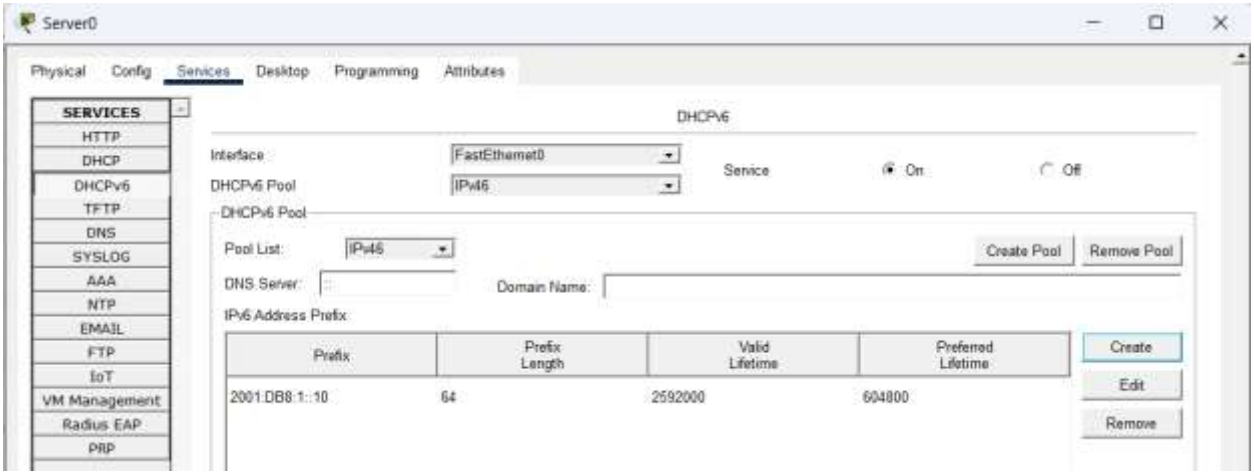
## IV. Conclusion

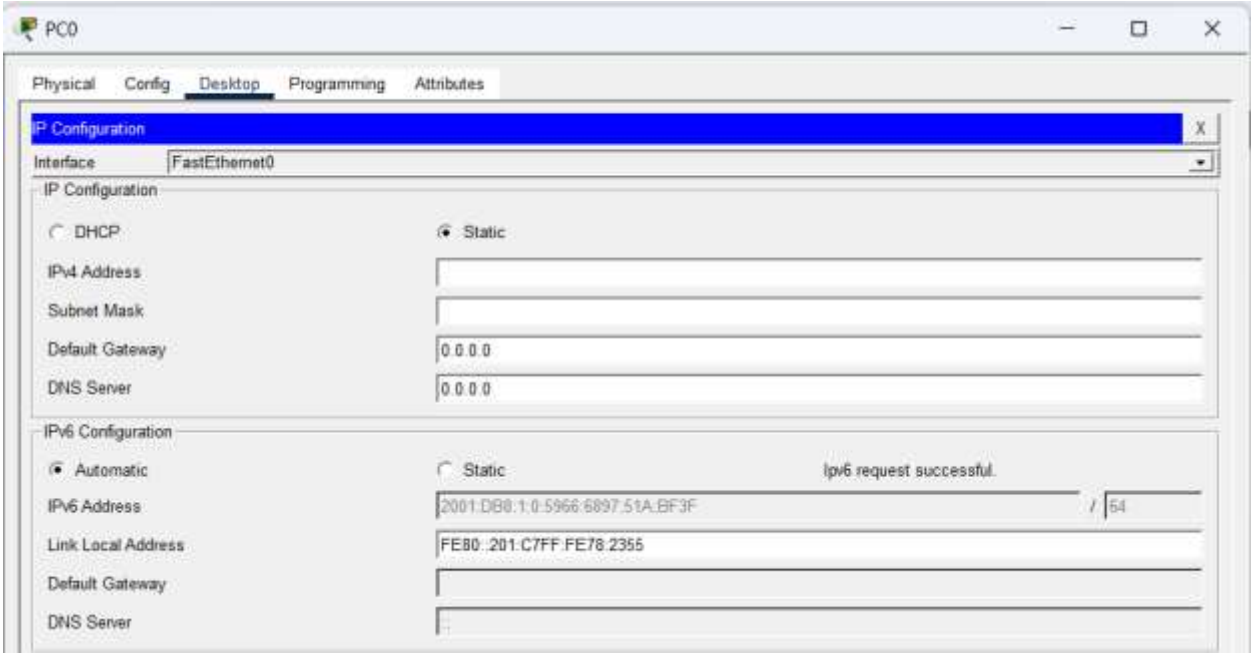
The activity showed how DHCPv6 can be configured in Cisco Packet Tracer to automatically assign IPv6 addresses to client computers. By enabling the DHCPv6 service on the server and setting up the IPv6 pool, the connected PCs were able to obtain IPv6 addresses without manually entering them. This proves that DHCPv6 helps make network configuration easier, faster, and more organized, especially when multiple devices need to be connected under the same IPv6 network.

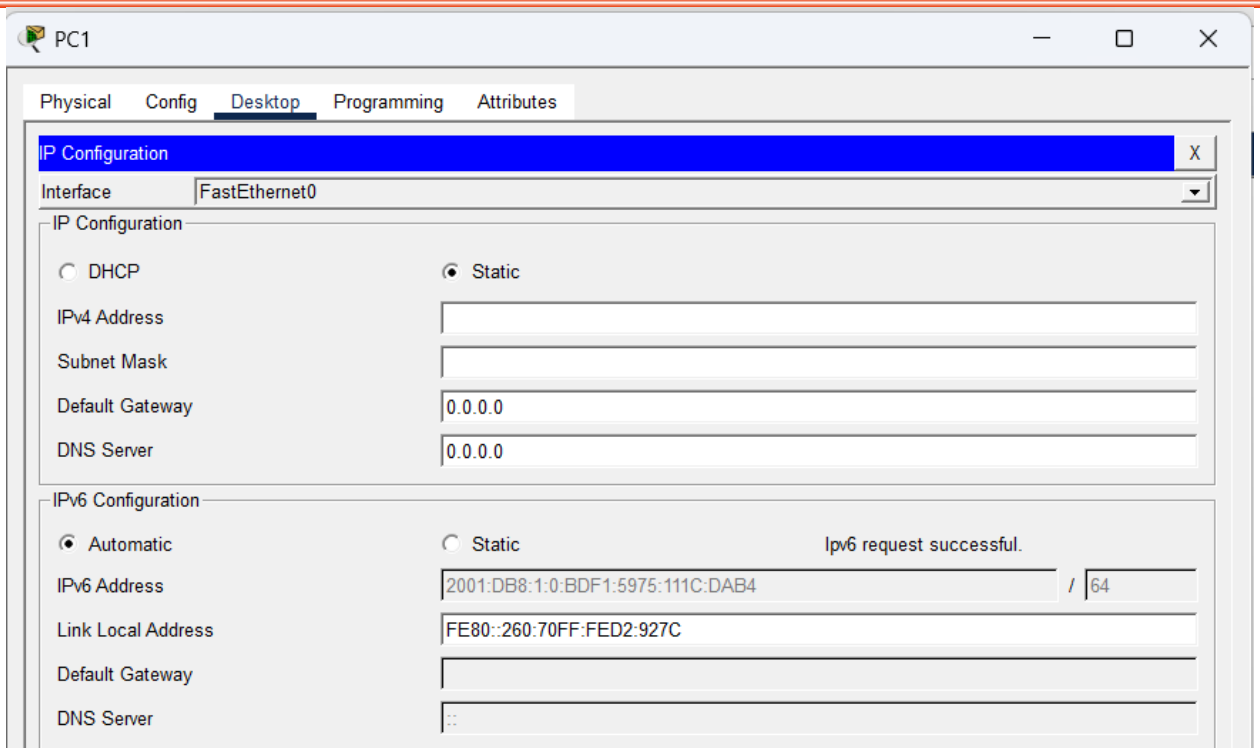
## V. Documentation

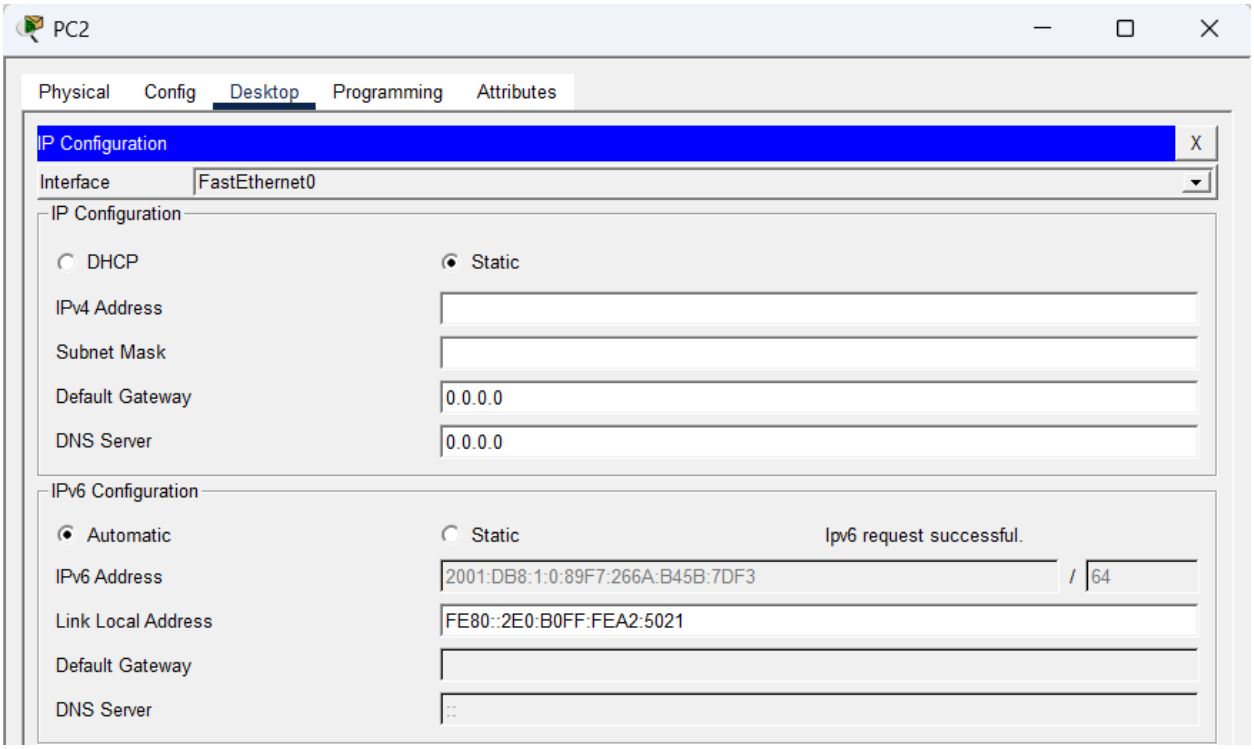



Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCPv6
Interface
FastEthernet0
DHCPv6 Pool
IPv46
Service
On
Off
DHCPv6 Pool
Pool List:
IPv46
Create Pool
Remove Pool
DNS Server:
Domain Name:
IPv6 Address Prefix
Prefix
Prefix Length
Valid Lifetime
Preferred Lifetime
2001:DB8:1::10
64
2592000
604800
Create
Edit
Remove


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
Ipv6 request successful.
IPv6 Address
2001:DB8:1:0:5966:6897:51A:BF3F
/ 64
Link Local Address
FE80::201:C7FF:FE78:2355
Default Gateway
DNS Server


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
Ipv6 request successful.
IPv6 Address
2001:DB8:1:0:BDF1:5975:111C:DAB4
/
64
Link Local Address
FE80::260:70FF:FED2:927C
Default Gateway
DNS Server
::


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
Ipv6 request successful.
IPv6 Address
2001:DB8:1:0:89F7:266A:B45B:7DF3
/
64
Link Local Address
FE80::2E0:B0FF:FEA2:5021
Default Gateway
DNS Server
::